TRANSISTOR LABS

FM-1 emulator

User Manual - Version 1.2.0

Free AU / VST3 / CLAP plugin for macOS, Windows and Linux

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1. Introduction

FM-1 emulator runs the firmware of the M-VAVE FM-1 pocket synth inside your DAW. It emulates the FM-1's processor, screen, keys, knobs, audio and MIDI, so firmwares behave as they do on the real device. You play it from your DAW's MIDI or with the mouse on its front panel.

FM-1 emulator is free. It is an unofficial product and is not affiliated with or endorsed by M-VAVE.

2. System Requirements

- **macOS:** macOS 12 Monterey or later, Mac with Apple silicon (M1 or later). AU, VST3 and CLAP.
- **Windows:** Windows 10 or 11, 64-bit (x64). VST3 and CLAP.
- **Linux:** 64-bit x86 distribution with X11 (e.g. Ubuntu 22.04 or later). VST3 and CLAP.
- A host that loads AU, VST3 or CLAP instruments (Logic, GarageBand, Ableton Live, Cubase, Reaper, Bitwig, FL Studio, Studio One...).
- An FM-1 firmware file (.fwsc). Firmware is not included: see chapter 4.

3. Installation

macOS: double-click *FM-1-emulator-1.2.0-macos.pkg* and follow the installer. It installs the AU, VST3 and CLAP into /Library/Audio/Plug-Ins.

macOS security warning. The installer is not notarised by Apple, so the first time you open it macOS says it cannot be opened because it is from an unidentified developer. To open it:

- Right-click (or Control-click) the .pkg and choose **Open**, then click **Open** again in the dialog.
- Or: try to open it once, then go to **System Preferences > Security & Privacy > General** (macOS 12) or **System Settings > Privacy & Security** (macOS 13 and later) and click **Open Anyway**.

Windows: run *FM-1-emulator-1.2.0-windows-x64-setup.exe*. It installs FM-1.vst3 into Common Files\VST3 and FM-1.clap into Common Files\CLAP. The VST3 needs the CLAP file to be installed too, so keep both. If Windows SmartScreen warns you, click **More info > Run anyway**.

Linux: unpack the .tar.gz and run *./install.sh*. It copies the plugins into ~/.vst3 and ~/.clap.

Plugins only: the -plugins archives contain the plugin files to copy by hand into your plugin folders.

Restart your DAW after installing. The plugin is listed as **FM-1 emulator** by **Transistor Labs**.

4. Getting a Firmware

The plugin needs an FM-1 firmware package (.fwsc) to make sound. Firmware is copyrighted by its authors and is not bundled. Community firmwares are available from their authors' websites; a list with links is on the fm1-emu project page (https://joakimeriksson.github.io/fm1-emu/). Download a .fwsc file and keep it somewhere you can find it.

Once loaded, the firmware is stored inside your DAW project, together with the patches it saves, so the project opens with the same sound next time.

5. The Plugin Window



At the top of the window is the control bar; below it is the FM-1 panel: MASTER, SELECT, PRESETS and ALGORITHM knobs on the left, the screen in the middle, KNOB 1-4 and the twelve function buttons on the right, OCT-/OCT+ and the 27-key keyboard (F3 to G5) at the bottom. Keys and buttons glow orange when the firmware lights their LEDs.

- **Load firmware / save...** opens a .fwsc firmware package or a shared .fm1save setup.
- **Export save...** saves the running firmware, its patches and settings as a .fm1save file you can share.
- **MIDI channel** moves all incoming notes to one channel. Most firmwares listen on a single channel (M-VAVE's own on 2): if your MIDI keyboard plays nothing while the panel keys do, set this to the firmware's channel.
- **Sync to host** (on by default) sends your DAW's tempo to the firmware as MIDI clock, with Start and Stop following the transport, so arpeggiators and sequencers play in time. It works with firmwares that follow MIDI clock.
- **CPU clock** sets the emulated chip's speed (240 MHz like the real one). Lower it if your computer can't keep up; raise it if a firmware runs out of time. Changing it restarts the firmware.
- **Lead** sets how far ahead of the host the firmware runs. Smaller is quicker to respond; go back up if "frames late" appears in the status bar.
- The **status** text shows the firmware name, speed and any late frames.
- **About** shows the version and credits.

6. Playing

- Play from your DAW's MIDI track or a MIDI keyboard.
- On the panel, click a key to play it; drag across the keys for a glissando.
- Drag a knob up or down, or use the mouse wheel over it.
- MASTER is the host parameter *Master*, so it can be automated.
- Buttons work while held down, as on the hardware.

An SMK-37 firmware package shows the SMK-37 panel instead, with its pads, faders and wheels.

7. Troubleshooting

- **No sound:** load a firmware first.
- **Panel keys play but the MIDI keyboard doesn't:** set MIDI channel to the firmware's channel (try 2, then 1). In Ableton, arm the track or set Monitor to In.
- **Not in time with the DAW:** check Sync to host is on and the firmware's own clock setting is set to external/MIDI if it has one.
- **Crackles or "frames late":** raise Lead, lower CPU clock, or increase your DAW's buffer size.
- **Plugin not listed:** restart the DAW and rescan plugins. On Windows the VST3 needs FM-1.clap in Common Files\CLAP.
- **"The firmware stopped":** the firmware used an instruction the emulator doesn't know yet. Try another firmware.

8. Credits, Licence and Contact

FM-1 emulator by Transistor Labs. Emulation core: fm1-emu by Joakim Eriksson, MIT licence. Built with nih-plug (ISC), clap-wrapper and the CLAP SDK (MIT), Apple's AudioUnitSDK (Apache-2.0) and Steinberg's VST3 SDK (MIT). Firmwares belong to their authors. M-VAVE and FM-1 are names of their owner; this is an unofficial emulator.

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